

HASSAN HAMAD

📍 San Francisco Bay Area, CA ✉ hmmhhamad@gmail.com 📞 7147338225

🌐 <https://www.hassanhamad.com>

🌐 <https://www.linkedin.com/in/hassan-hamad-338a7513b/>

PROFILE

An AI researcher with a passion for developing intelligent systems and understanding the fundamentals that drive them. I am currently working on AI agents that power next-generation solutions and products. In my free time, I enjoy following the latest research literature and reproducing new results from papers that I find interesting. Check out my technical blog [here](#)!

EDUCATION

- 📅 2019/08–2024/12 **PhD in Electrical and Computer Engineering**
University of Southern California (USC) 📍 Los Angeles, CA
» Focus on Efficient Training of Deep Learning Models from Compute and Data Side.
» Cumulative GPA: 4.0/4.0
- 📅 2016/09–2019/05 **M.S. in Communications Engineering**
Technical University of Munich (TUM) 📍 Munich, Germany
» Focus on Information Theory and Wireless Communications
» Cumulative GPA: 3.9/4.0
- 📅 2012/09–2016/06 **B.E. in Computer and Communications Engineering**
Notre Dame University - Louaize 📍 Zouk Mosbeh, Lebanon
» Focus on Algorithms, Digital Coding and Wireless Communications
» Cumulative GPA: 3.9/4.0

WORK EXPERIENCE

- 📅 2025/01–Present **Applied Scientist**
Amazon, AWS AI Fundamentals 📍 Santa Clara, CA
» Developing domain-specialized AI agents to automate and optimize complex workflows.
- 📅 2024/05–2024/08 **Applied Scientist Intern**
Amazon, Alexa Shopping 📍 Seattle, WA
» Utilized synthetic data to improve LLM function-calling in dialogue systems.
- 📅 2023/05–2023/08 **Quantitative Machine Learning Intern**
Bloomberg 📍 New York, NY
» Enhanced information extraction capabilities through LLMs in financial data.
- 📅 2019/02–2019/06 **Working Student - Machine Learning**
Fraunhofer 📍 Munich, Germany
» Explored the use of RNNs for predicting latency of a V2V communication link.
- 📅 2017/05–2018/05 **Working Student - LTE physical layer system engineer**
Intel 📍 Munich, Germany
» Worked on a physical layer concept to implement the V2X feature from 3GPP.

TEACHING EXPERIENCE

- 📅 2018/10–2019/02 **Lab Assistant for Communications Lab**
Technical University of Munich (TUM) 📍 Munich, Germany

» Introduction to communication systems based on experiments and computer simulations.

SOFTWARE SKILLS

Python	● ● ● ● ● ● ● ● ●
PyTorch	● ● ● ● ● ● ● ● ●
C++	● ● ● ● ● ● ● ● ●
CUDA	● ● ● ● ● ● ● ● ●
Tensorflow	● ● ● ● ● ● ● ● ●
Linux	● ● ● ● ● ● ● ● ●
AWS	● ● ● ● ● ● ● ● ●
Docker	● ● ● ● ● ● ● ● ●

SELECTED PUBLICATIONS

- » [ToolCritic: Detecting and Correcting Tool-Use Errors in Dialogue Systems](#)
👤 **H. Hamad**, Y. Xu, L. Zhao, W. Yan, N. Gyanchandani
📖 *arXiv e-prints*, *arXiv*: 2510.17052
- » [Bitwidth-Specific Logarithmic Arithmetic for Future Hardware-Accelerated Training](#)
👤 **H. Hamad**, Y. Qiu, P.A. Beerel, K.M. Chugg
📖 *arXiv preprint arXiv:2510.17058*
- » [FIRE: A Dataset for Financial Relation Extraction](#)
👤 **H. Hamad**, A. Thakur, S. Pulikodan, N. Koller and K.M. Chugg.
📖 *Findings of the 2024 Annual Conference of the North American Chapter of the ACL (NAACL)*.
- » [Approximation Capabilities of Neural Networks using Morphological Perceptrons and Generalizations](#)
👤 W. Chang, **H. Hamad** and K.M. Chugg.
📖 In *2022 Asilomar Conference on Signals, Systems, and Computers*.
- » [Joint State Sensing and Communication over Memoryless Multiple Access Channels](#)
👤 M. Kobayashi, **H. Hamad**, G. Kramer and G. Caire.
📖 In *2019 IEEE International Symposium on Information Theory (ISIT)*.
- » [Performance Analysis of Convolutional Codes over the Bernoulli-Gaussian Impulsive Noise Channel](#)
👤 **H. Hamad** and G. M. Kraidy.
📖 In *2017 15th Canadian Workshop on Information Theory (CWIT)*.

ACADEMIC ACHIEVEMENTS

📅 2019/08–2020/08	Annenberg Fellowship recipient University of Southern California (USC)	📍 Los Angeles, CA
📅 2020/08–present	SLK America Fellowship recipient V-Labs (in partnership with SLK Software)	📍 Los Angeles, CA

LANGUAGES

Arabic (native) English (fluent) German (basic)